Exam 2019 solutions

B—bookwork, E—new example, T—new theory

1.

a)

(i) Note that $0 < z < 1$.

$$F(Z) = P(Z \le z) = P(XY \le z)$$

2E

$$F(Z) = P\left(Y \le \frac{Z}{X}\right) = \int_0^z \int_0^1 dy\, dx + \int_z^1 \int_0^{\frac{z}{x}} dy\, dx$$

2E

$$= z + \int_z^1 \frac{z}{x} dx = z + z\,[\ln(1) - \ln(z)] = z - z\ln(z)$$

$$f(z) = 1 - (\ln(z) + 1) = -\ln(z), \quad 0 < z < 1$$

2E

(ii) $a = 1$ in the following.

$$F_Z(z) = P\left\{\frac{X}{Y} \le z\right\} = P\{X \le zY\}$$

$z < 1$

$$F_Z(z) = P\{X \le zY\}$$
$$= \int_0^a \int_0^{zy} \frac{1}{a} \cdot \frac{1}{a}\, dx\, dy = \frac{z}{2}, \quad z \le 1$$

2E

$z \ge 1$

$$F_Z(z) = P\{X \le zY\}$$
$$= 1 - \int_0^a \int_0^{x/z} \frac{1}{a} \cdot \frac{1}{a}\, dy\, dx$$
$$= 1 - \int_0^1 \frac{x}{z}\, dx = 1 - \frac{1}{2z} \quad z > 1$$

2E

$$f_Z(z) = \begin{cases} \frac{1}{2}, & z \le 1 \\ \frac{1}{2z^2}, & z > 1 \end{cases}$$

2E

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(iii)

$$F_Z(z) = P(Z \le z) = P\left\{\frac{Y}{X+Y} \le z\right\}$$
$$= P\left\{\frac{X}{Y} \ge \frac{1}{z} - 1\right\} = 1 - P\left(\frac{X}{Y} \le \frac{1-z}{z}\right)$$
$$= \begin{cases} \frac{1}{2}\left(\frac{z}{1-z}\right), & 0 < z \le 1/2 \\ 1 - \frac{1}{2}\left(\frac{1-z}{z}\right), & 1/2 < z < 1 \end{cases}$$

[2E] [2E]

$$f_Z(z) = \begin{cases} \frac{1}{2(1-z)^2}, & 0 < z \le 1/2 \\ \frac{1}{2z^2}, & 1/2 < z < 1 \end{cases}$$

[2E]

b)

$$E[X] = \int_0^\infty x f(x) dx = \int_0^\infty \int_0^x 1 \cdot dy\, f(x) dx$$ [2T]

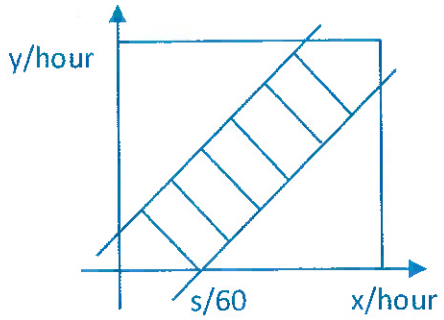
$$= \int_0^\infty \int_0^x f(x) dy\, dx = \int_0^\infty \int_y^\infty f(x) dx\, dy$$ [3T]

$$= \int_0^\infty P(X \ge y) dy = \int_0^\infty P(X \ge x) dx$$ [2T]

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2.

a)



[2E]

See the unit square in the figure, the origin is 5:00pm, x axis (unit: hour) represents the time Alice arrive and y axis represents the time Bob arrives. [3E]

So, "each is prepared to wait s/60 hours before leaving and they meet each other" can be mathematically described as

$\left\{|x-y| \le \frac{s}{60}\right\}$ (see the shadow area in the figure)

[3E]

$$P\left(|x-y| \le \frac{s}{60}\right) = \iint_{|x-y| \le \frac{s}{60}} f(x,y)\,dx\,dy = 1 - 4 \times \left(\frac{1}{2} - \frac{s}{60}\right)^2$$

where (x,y) is a two-dimensional variable stands for the time Alice and Bob arrive, and its density function is $f(x,y) = \begin{cases} 4, & 0 \le x \le 0.5, 0 \le y \le 0.5 \\ 0, & \text{otherwise} \end{cases}$ [2E]

Finally, $1 - 4 \times \left(\frac{1}{2} - \frac{s}{60}\right)^2 \ge \frac{3}{4} \Rightarrow s \ge 15 \text{ min}$ [3E]

b) The joint distribution of n observations (where $\bar{x} = 6$ is the empirical mean) [3E]

$$f(X,\beta) = \frac{\beta^{n\alpha}(x_1 \dots x_n)^{(\alpha-1)} e^{-n\beta\bar{x}}}{\Gamma^n(\alpha)}, \quad x > 0, \alpha = 3, \beta > 0$$

Take partial derivative: [3E]

$$\frac{\partial f(X,\beta)}{\partial \beta} = \frac{n\alpha\beta^{n\alpha-1}(x_1 \dots x_n)^{(\alpha-1)} e^{-n\beta\bar{x}}}{\Gamma^n(\alpha)} - n\bar{x}\frac{\beta^{n\alpha}(x_1 \dots x_n)^{(\alpha-1)} e^{-n\beta\bar{x}}}{\Gamma^n(\alpha)}$$

[3E]

$$= \left(\frac{n\alpha}{\beta} - n\bar{x}\right) f(X,\beta)$$

Thus, letting it be 0, we have $\beta = \frac{\alpha}{\bar{x}} = \frac{4}{6} = \frac{2}{3}$ [3E]

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3. a) (i) $E[X(t)] = t + 0.5 \sin(\pi t)$ [4B]

(ii) $E[X(t_1)X(t_2)] = 2t_1 t_2 + 0.5 \sin \pi t_1 \sin \pi t_2$ [4B]

(iii) Not stationary [2B]

b) Firstly note that the arrival rate $\lambda = 0.2$ student/minute. [1E]

i) The probability that k students are being serviced at a given (random) time, is exactly the probability that k students arrived within the last 10 minutes, but this probability is Poisson distributed, hence [2E]

$$P(X = k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t} = \frac{2^k}{k!} e^{-2}, \quad k = 0,1,2, \ldots$$ [2E]

ii) The second case can be thought of random selection of a Poisson process, which gives rise to two Poisson processes, each with arrival rate $\lambda = 0.1$. This yields two queues with difference service times. The number of customers served is therefore the sum of the two queues, i.e., $k = k_1 + k_2$, where k_1 is Poisson distributed with parameter 1 and k_2 is Poisson distributed with parameter 2. [3E] [2E]

We know that the sum of two Poisson distributions is again Poisson with parameter equal to the sum of the individual distributions' parameters, i.e., the number of students is Poisson distributed (with parameter 3) as [2E]

$$P(X = k) = \frac{3^k}{k!} e^{-3}, \quad k = 0,1,2, \ldots$$ [3E]

4.

a) *The transition probability matrix is given by*

$$\mathbf{P} = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 \end{pmatrix}$$

[3E]

The limiting probabilities are obtained by solving the equations

$$\begin{aligned} \pi_0 &= \pi_1 + \frac{1}{2}\pi_2 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4 \\ \pi_1 &= \frac{1}{2}\pi_2 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4 \\ \pi_2 &= \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4 \\ \pi_3 &= \frac{1}{4}\pi_4 \\ \pi_4 &= \pi_0 \end{aligned}$$

[3E]

together with the normalization equation

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$$

[3E]

We obtain

$$\begin{aligned} \pi_4 &= \pi_0 \\ \pi_3 &= \frac{1}{4}\pi_4 \Rightarrow \pi_3 = \frac{1}{4}\pi_0 \\ \pi_2 &= \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4 \Rightarrow \pi_2 = \frac{1}{3}\left(\frac{1}{4}\pi_0\right) + \frac{1}{4}\pi_0 = \frac{1}{3}\pi_0 \\ \pi_1 &= \frac{1}{2}\pi_2 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4 \Rightarrow \pi_1 = \frac{1}{2}\left(\frac{1}{3}\pi_0\right) + \frac{1}{3}\left(\frac{1}{4}\pi_0\right) + \frac{1}{4}\pi_0 = \frac{1}{2}\pi_0 \end{aligned}$$

[3E]

therefore

$$\pi_0\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + 1\right) = 1 \Longrightarrow \pi_0 = \frac{12}{37}$$

and we get

$$\begin{aligned} \pi_0 &= \frac{12}{37} = 0.324 \\ \pi_1 &= \frac{6}{37} = 0.162 \\ \pi_2 &= \frac{4}{37} = 0.108 \\ \pi_3 &= \frac{3}{37} = 0.081 \\ \pi_4 &= \frac{12}{37} = 0.324. \end{aligned}$$

[3E]

EE4-10

b)

i) Of the $2n$ steps, the chain must goes left for n steps, and right for n.
Therefore,

$$P\{X_{2n}=0\}=\left(\frac{1}{2}\right)^{2n}\binom{2n}{n}$$ [4E]

ii)

$$\sum_n P\{X_{2n}=0\}=\sum_{n\geq 1}\left(\frac{1}{2}\right)^{2n}\binom{2n}{n}=\sum_{n\geq 1}\left(\frac{1}{2}\right)^{2n}\frac{(2n)!}{n!\,n!}$$
$$=\sum_{n\geq 1}\left(\frac{1}{2}\right)^{2n}2^{2n}\frac{1}{\sqrt{\pi n}}=\sum_{n\geq 1}\frac{1}{\sqrt{\pi n}}=\frac{1}{\sqrt{\pi}}\zeta\left(\frac{1}{2}\right)=\infty$$ [3E]

because the sum diverges. We therefore conclude that the origin is a recurrent state. [3T]

Here $\zeta(s)$ is the Riemann zeta function, which is well-known to converge for $s>1$ but diverge otherwise.

PS. The famous Riemann hypothesis, considered one of the greatest unsolved problems in mathematics, asserts that any non-trivial zero s of $\zeta(s)$ on the complex plane has Re(s) = 1/2.